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Local government and climate literacy: exploring community perspectives in Ghana's coastal zones

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ABSTRACT

Effective climate change education by District Assemblies is crucial for fostering community adaptation. A cross-sectional survey design with a quantitative method was adopted to examine coastal residents' perceptions of District Assemblies in climate change education. A sample size of 420 was proportionally drawn from Akwidaa, Anlo Beach, Ampenyi and Gomoa Fetteh in the Western and Central Regions of Ghana, respectively. The findings indicated that 53% of coastal residents strongly agreed that District Assemblies play a critical role in climate change education. Yet, 72% believed these assemblies have not adequately educated their communities in the past 5 years. In addition, 45.5% rated their access to climate change information from District Assemblies as very poor. There was a statistically significant influence of the community on respondents' satisfaction with public climate change education ($\beta = 0.400$; $p = 0.000$). Furthermore, the results revealed that the type of communicator, the information channel employed, trust in DAs, and attention to climate change information influenced climate education effectiveness. The study underscores the need for enhancing structural and systemic factors within communities in climate change education than individual socio-demographic factors to improve climate literacy and foster effective environmental stewardship in Ghana's coastal regions.

Key policy highlights

- District Assemblies should enhance climate change education by focusing on community-centric programs that leverage local knowledge, making content more accessible and relevant to coastal communities.
- Trusted communicators and accessible media should be used to share climate change information.
- To build trust and enhance climate literacy, Assemblies must collaborate with local leaders, NGOs and educational institutions, co-creating initiatives that strengthen community resilience and environmental stewardship.

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Introduction

The recognition of local governments as essential actors in climate action has grown significantly, particularly with the increasing decentralization of governance structures and the shifting focus toward bottom-up approaches to environmental sustainability (Bulkeley & Betsill, 2013). Positioned between national policies and local communities, local governments are well-suited to enhance climate literacy, empowering citizens to understand and adapt to climate risks (Monroe et al., 2019; IPCC, 2021). Globally, local governments play

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a critical role in advancing climate literacy, ensuring that communities are well-informed about climate risks and equipped to participate in adaptation strategies (Satterthwaite, 2016). While national governments create climate policies, implementation often occurs locally. This proximity allows local governments to tailor climate education to regional needs, as seen in cities like Copenhagen and New York, where climate literacy is integrated into services, education and planning (Rosenzweig et al., 2018). In the Netherlands, local governments ensure citizens and policymakers are well-informed about climate risks and adaptation strategies (Biesbroek et al., 2009). However, in developing regions, particularly sub-Saharan Africa's coastal zones – local governments often face challenges, including limited local autonomy and funding (UNESCO, 2019; Hughes, 2017), weak institutional capacity, and political resistance (UNDP, 2020; Moser & Berkes, 2019), leading to low climate literacy levels (Satterthwaite, 2016; IPCC, 2021).

Africa's coastal zones, including Ghana, are highly vulnerable to climate impacts, threatening livelihoods, infrastructure and food security. Local governments in these regions are tasked with balancing the pressures of development with the need to protect coastal communities from environmental hazards. However, their ability to respond effectively often depends on the level of climate literacy within both the government and the communities they serve (Biesbroek et al., 2010). In the Global South, local governments have begun to prioritize community engagement and education in climate change discourse, recognizing that successful adaptation strategies require both top-down policy implementation and grassroots participation (Spires et al., 2014). This involves participatory governance where residents are involved in climate adaptation decisions, fostering collaboration between local governments and communities (Meadowcroft, 2009). Local governments must prioritize vulnerable populations, as they are often the most affected by climate change (Schlosberg, 2012). In urban areas like Dhaka and Lagos, local governments work to improve climate literacy in informal settlements, where residents face heightened risks from flooding and other climate-related hazards (Roy et al., 2018). Traditional knowledge systems also contribute to climate adaptation. Local governments that integrate traditional knowledge with scientific approaches can achieve more sustainable outcomes. For instance, in the Arctic regions of Canada and Alaska, local governments have collaborated with indigenous communities to develop climate literacy programs that combine scientific knowledge with traditional ecological knowledge (TEK), leading to more culturally relevant and effective adaptation strategies (Ford et al., 2020). In addition, partnerships between local governments, civil society and international organizations are crucial for advancing climate literacy. NGOs and development agencies, such as USAID, provide funding and expertise to support climate education. In this way, local governments can serve as facilitators of climate education, while external partners provide the necessary resources and technical support (Rosenzweig et al., 2018).

Despite growing concerns about climate change impacts in Ghana's coastal zones, limited research has examined the intersection of local government engagement and public climate literacy. Existing studies tend to focus on the broader role of District Assemblies in community development, while overlooking community perspectives on the effectiveness, consistency and trustworthiness of these efforts, revealing a critical gap in understanding local governance's role in climate education. This study fills this gap by examining District Assemblies' role in climate change education, the availability and access to and satisfaction with climate information from District Assemblies, the influence of socio-demographic backgrounds on respondents' satisfaction with climate change education and, finally, factors that determine the effectiveness of climate education. Understanding this is essential for enhancing local governance strategies and tailoring climate education programs that resonate with coastal communities in Ghana, ultimately fostering more effective environmental stewardship. Insights from Ghana's coastal zones can inform policies in other vulnerable regions, offering context-specific strategies for integrating climate education into local-level action. This study contributes to global climate governance by promoting informed, locally-driven climate solutions.

The rest of the paper is structured as follows. Section 2 explains the methodology employed to achieve the study's objectives. Section 3 presents the results. Section 4 details the discussion of the findings. Section 5, the final section outlines the conclusion and implications for policy.

Literature review

Education and climate literacy

Public awareness and education are pivotal in climate literacy, empowering societies to make informed choices and advocate for policies to adapt to climate change impacts. Although various organizations and individuals have long been active at the crossroads of education and environmental issues, education as a whole has seldom been given priority as a solution to the climate crisis. Education needs to be more prominently featured in global climate discussions and national adaptation strategies, and the global education sector must rally for climate action (Ehlers et al., 2021). In the context of this study, informal education platforms – such as media, community programs and digital campaigns – has gained recognition for their ability to reach diverse audiences and bridge gaps in formal education. Corner et al. (2018) argue that these platforms play a crucial role in making climate science more accessible to the general public, especially in cases where formal education may be lacking or where there is a need for continuous learning beyond the classroom. Informal education initiatives can engage individuals across different age groups and social backgrounds, facilitating the broader public engagement with climate issues. For example, media campaigns that leverage social media, documentaries and news outlets can disseminate climate-related information widely and rapidly, helping to raise awareness and inspire action. In many developing countries, including Ghana, community information centres have arisen as a prominent media platform for enlightening the populace about climate change (Adjei Baffour et al., 2024). The authors contend that rather than functioning as mechanisms for exploiting communities under the pretext of promoting traditional medication, indigenous platforms may be strategically harnessed by local governments to enhance public awareness of the benefits of proactive engagement and the potential consequences of inaction.

Despite the recognized importance of climate literacy, challenges such as misinformation, political polarization and cognitive biases can hinder public understanding and acceptance of climate science. The proliferation of misinformation, particularly through digital media, can create confusion and scepticism, undermining efforts to build climate literacy (Moser & Dilling, 2011). Addressing these challenges requires educators and communicators to develop innovative strategies that not only disseminate accurate scientific information but also engage audiences in meaningful dialogue and critical reflection. Participatory approaches that involve communities in climate education initiatives can enhance understanding and foster a sense of ownership and responsibility toward climate action (O'Brien et al., 2010). Citizen participation encompasses the tools and methods that enable individuals to actively participate in the planning, implementation and assessment of climate change adaptation efforts (Cid & Lerner, 2023).

Intersections between local governance and environmental education

Global attempts to deal with climate change have yielded mixed results, with the 'Nationally Determined Contributions (NDCs)' provided under the 2015 Paris Agreement proving inadequate to lessen global warming to less than 2°C above pre-industrial levels, and falling even further short of the aspirational 1.5°C target. Despite this shortfall in ambition from national governments, municipal, regional and provincial authorities are actively working to tackle climate change within their own jurisdictions (Dale et al., 2020). According to Hermelin & Gustafsson (2024), effective local governance enhances the implementation of environmental education programs by fostering community participation and ensuring that educational initiatives are contextually relevant. Decentralized governance structures, such as local councils and community-based organizations, are better positioned to tailor environmental education to the specific need of their communities, thereby increasing the efficacy of such programs (Moseley, 2018). Through collaboration with non-governmental organizations and academic institutions, local governments can access a broader range of expertise and resources, which can enrich the content and delivery of climate education programs (Wheeler, 2008). Furthermore, through public workshops and community meetings, local governments can provide residents with practical knowledge about adapting to climate impacts (Picketts et al., 2012). Local governments often serve as models for climate action by implementing sustainable practices within their operations, thereby demonstrating the feasibility and benefits of such measures (Evans et al., 2017). Lee and Williams (2020) argue that when local authorities prioritize environmental education, they empower communities to take proactive steps in conserving local ecosystems and managing natural resources sustainably. This symbiotic relationship between local governance

and environmental education also fosters a sense of environmental responsibility among citizens, which is essential for the sustained success of environmental initiatives (Powers, 2017).

In Ghana, local governments facilitate climate change education through community-based programs and workshops that promote sustainable practices (Mensah & Amaning, 2015). For instance, initiatives such as 'the Climate Change Adaptation in Northern Ghana Enhanced (CHANGE) project' have been instrumental in enhancing climate literacy among local communities (Dzodzi, 2017). By working closely with traditional leaders to leverage indigenous knowledge systems in climate change education, local governments foster a more holistic understanding of environmental stewardship (Owusu-Daaku & Diko, 2018). However, the effectiveness of the Assemblies often hinges on public trust and engagement. Trust in local authorities is crucial, as it influences community participation in climate education programs (Boateng, 2019). Mensah et al. (2021) reveal that while some communities perceive local governments as essential partners in climate action, others view them as inefficient and corrupt, undermining educational initiatives. This study uniquely contributes to the literature by examining residents' views from Ghana's coastal zones.

Theoretical frameworks

Social learning theory

'Social Learning Theory' (SLT), developed by Albert Bandura, is a pivotal model for comprehending how people and communities learn via observation, imitation and modelling (Bandura, 1977). This theory highlights the significance of social interactions and the environment in influencing behaviour. As local governments are increasingly tasked with implementing climate adaptation strategies, SLT offers valuable insights into how these strategies can be effectively communicated and adopted within communities. Through processes like participatory planning and community workshops, local governments can facilitate social learning by creating platforms where citizens can share experiences and learn from each other's successes and failures (Reed et al., 2010). This collective learning process is essential in building climate resilience, as it fosters a shared understanding of climate risks and encourages the adoption of sustainable practices (Pahl-Wostl et al., 2007). For instance, when local governments engage communities in climate action, they enable citizens to observe and model behaviours that contribute to environmental sustainability (Muro & Jeffrey, 2008). These behaviours can then spread through social networks, leading to a broader cultural shift toward sustainability (Bandura, 2001).

The theory of planned Behaviour

Icek Ajzen created the 'Theory of Planned Behaviour (TPB)' in 1991 as a psychological framework explaining how attitudes, subjective norms and perceived behavioural control influence individual behaviour (Ajzen, 1991). In this study, TPB provides a valuable lens for understanding how decision-makers and the public might engage in climate-positive actions. If government officials have a positive attitude towards climate action, believe that their constituents expect them to address climate issues, and feel confident in their ability to implement effective policies, they tend to prioritize climate literacy efforts (Fielding et al., 2008). Governance structures that actively engage communities in climate education can shape attitudes and norms, thus encouraging climate-positive behaviours (Ockwell et al., 2009; Ajzen, 2002). TPB suggests that increasing public knowledge about climate change can positively influence attitudes and perceived control, leading to higher engagement in climate-friendly practices (Ajzen, 1991). If local authorities implement educational campaigns that emphasize adaptation and demonstrate practical ways to do so, they can foster a community-wide shift towards sustainable behaviours (Kollmuss & Agyeman, 2002).

Methodology

Study area

The study took place in four chosen fishing communities along Ghana's western and central coasts (Figure 1): Anlo Beach in the Shama District (5.0252°N, -1.6144°W), Akwidaa in the Ahanta West District (4.7603°N,

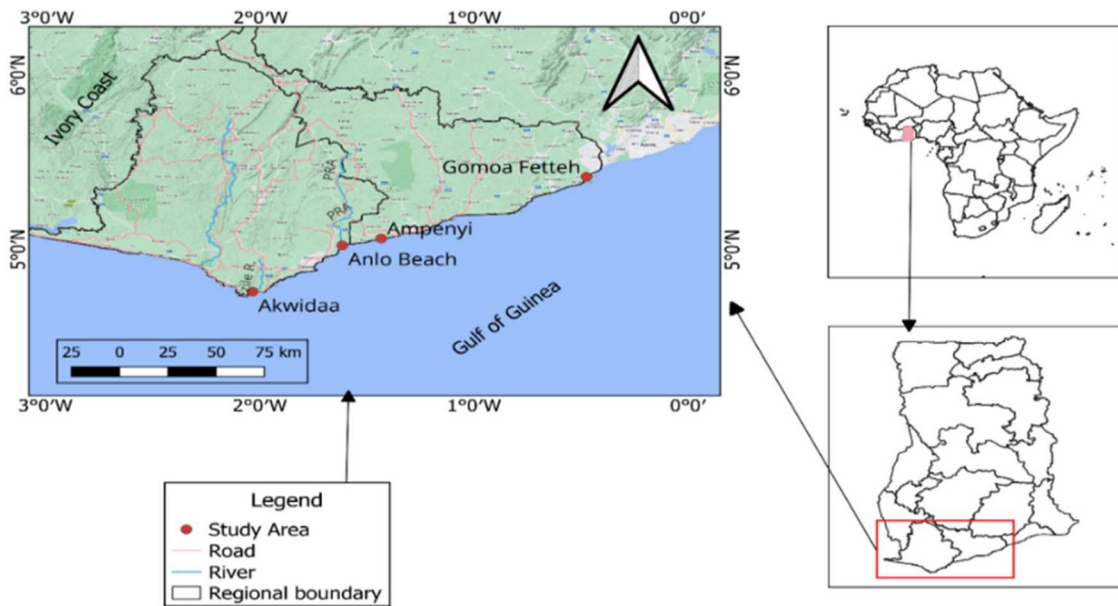


Figure 1. Map of the study area, indicating the four sampling communities.

–2.0341°W), Ampenyi in the Komenda Edina Eguafo Abirem District (5.0667°N, –1.4333°W) and Gomoa Fetteh in the Gomoa East District (5.4200°N, –0.4738°W). Except for Gomoa Fetteh, these communities share the geographical trait of being located between two water bodies. Anlo Beach lies between the sea and the Pra River estuary, Akwidaa is flanked by the sea and the Ezile River estuary and Ampenyi is nestled between the sea and the Brenu Lagoon. These communities were selected for their high vulnerability to climate impacts, with their coastal locations making them susceptible to persistent coastal inundations. Their dependence on fishing and farming as major sources of livelihoods, heightens their sensitivity to environmental changes, necessitating effective climate education (Adjei Baffour et al., 2024). These challenges are further exacerbated by gaps in knowledge (Letsyo, 2025), which could be caused by the absence of climate education. Furthermore, these communities were selected due to limited research focus. Although these areas face significant climatic challenges, scholarly attention has predominantly centred on more urbanized coastal zones, leaving these areas relatively underexplored in existing climate-related research.

Research design and sampling

A cross-sectional survey design, appropriate for capturing the perceptions of different communities at one particular moment in time (Creswell, 2014), was utilized in the investigation. This design is appropriate for the study as it allows for identification of trends and gaps, offering empirical insights into the effectiveness, accessibility and satisfaction with climate education. The quantitative approach allowed for collecting and analyzing numerical data, providing a broad understanding of the communities' perceptions (Fowler, 2014). The population for this research involved residents of selected coastal districts in the Central and Western Regions of Ghana. The study employed the stratified random sampling strategy to guarantee representation from various demographic categories. The strata were based on the communities within each district, and households were randomly selected within each community. The sample size was determined using Yamane's formula for large populations:

$$n_0 = \frac{N}{1 + N(e)^2} \quad (1)$$

where (N) is the total population and (e) is the precision level. With a 95% confidence level, the margin of error (e) is 0.05. Hence, from a population of 24,855,

$$n_0 = \frac{24855}{1 + 24855(0.05)^2} = 394$$

To account for potential non-responses, the investigator added 26 to the sample size, making a final total sample of 420 respondents.

Data collection, processing and analysis

A standardized survey was utilized to gather data from respondents. The investigator developed the questionnaires based on existing literature and pre-tested them to ensure reliability and validity. Items on the instrument were measured on a yes or no, strongly disagree to strongly agree, and very dissatisfied to very satisfied. Alternatively, a five-item satisfaction scale statement different from the former was transformed into one continuous satisfaction variable for multiple linear regression analysis. Similarly, the effectiveness of climate education question items was measured on a binary scale of 'yes' or 'no'. The researcher, along with trained assistants, collected data using self-administered questionnaires. These assistants received training on the study's objectives, ethical considerations and proper administration of the questionnaire. The data collected were serialized, coded and entered into statistical software. After preliminary analysis to check for any possible outliers and the assumption of parametric tests had led to satisfactory results, the data were then analysed using 'Statistical Package for the Social Sciences (SPSS) software'. While the descriptive statistics, mainly frequencies and percentages were used to summarize the data, inferential statistics, primarily regression analyses, were used to establish associations and test hypotheses regarding factors influencing the effectiveness of climate change education and satisfaction level with climate change education.

Results

Socio-demographic information of respondents

Table 1 illustrates respondents' socio-demographic information. In terms of gender, 49% of females and 51% of males took part in the study. Age groups indicated that 19% were within the age range of 21–30 years, 30%

Table 1. Respondents' socio-demographic background.

Variable	Group	Frequency (N)	Percentage (%)
Gender	Male	210	51
	Female	205	49
	Total	415	100
Age	21–30	80	19
	31–40	124	30
	41–50	81	20
	51–60	63	15
	60+	67	16
	Total	415	100
Community	Akwidaa	141	34
	Anlo Beach	76	18
	Ampenyi	68	16
	Gomoa Fetteh	130	31
	Total	415	100
Educational Level	No formal education	152	37
	Primary education	117	28
	Middle school	86	21
	Secondary education	48	12
	Colleges of education	4	1
	University education	8	2
	Total	415	100

Source: Field survey (2022).

within 31–40 years, 20% within 41–50 years, 15% within 51–60 years, and 16% within the age group of 60 and above years old. Regarding the community, while Akwidaa and Gomoa Fetteh constituted the significant majority with 34% in Akwidaa and 31% in Gomoa Fetteh, Anlo Beach and Ampenyi accounted for 18% and 16%, respectively. The educational level showed a decreasing trajectory with increasing levels of education with the majority of respondents falling into the lower educational category of no formal education (37%) and primary education (28%). In addition, 21% had middle school certificates, 12% with secondary school education, 2% with university education and 1% with colleges of education.

District assemblies' involvement in climate education

The study examined District Assemblies' involvement in educating communities about the climate crisis from the perspective of coastal residents. The results in Figure 2 indicated that the majority (53%) of coastal residents recognized the critical role of District Assemblies in public education on climate change by strongly agreeing that District Assemblies are involved in public education on climate change. In addition, 20% agreed that District Assemblies participate in educating communities about climate change. While 9.4% of respondents were neutral in this context, 12% and 5.5% disagreed and strongly disagreed, respectively, with the District Assemblies' participation/role in public climate change education. In testing the difference in DAs' involvement in climate education among different districts, Chi Square test revealed no significant difference in Assemblies' involvement in climate education among the different districts ($\chi^2 = 20.66$; $p = 0.056$).

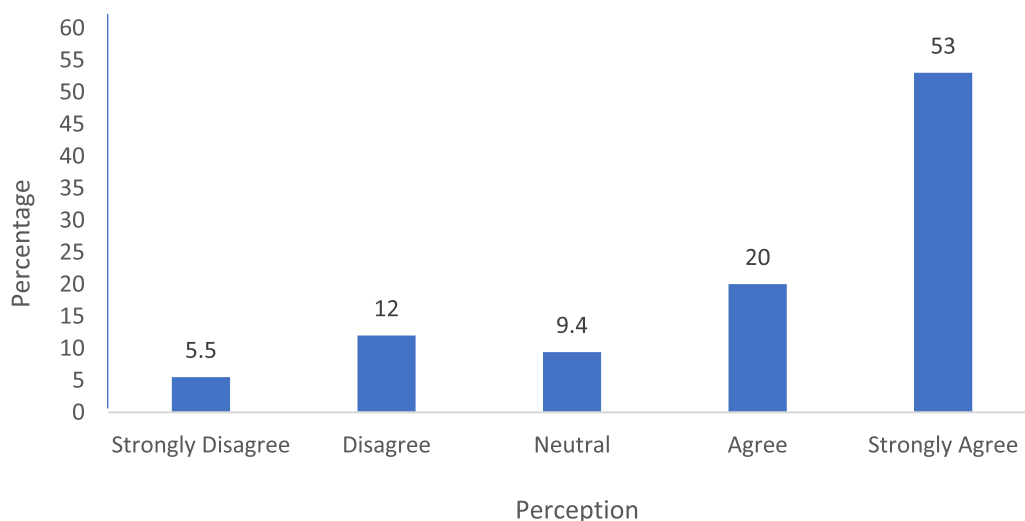


Figure 2. District Assemblies in climate change education.

DAs in climate education over the last 5 years

The study sought respondents' views about the role of the District Assemblies in educating coastal communities about the climate crisis in the last 5 years. The results in Figure 3 revealed that most (72%) of the respondents were of the recognition that the District Assemblies were not educating their constituent communities on the issues regarding climate change over the past half-decade. Only 28% of respondents perceived the Assemblies in educating their communities on climate change issues over the last 5 years.

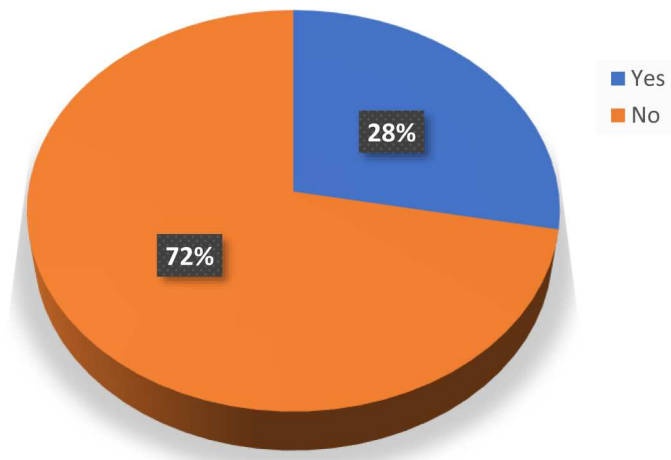


Figure 3. District Assemblies in climate change education over the past 5 years.

Availability and access to climate change information from the district assemblies

This part of the study examined the availability and access to climate change information from District Assemblies of the study's communities. The study, as shown in [Figure 4](#), uncovered that while the majority (71%) rated their availability and access to climate change information from the District Assemblies as poor and very poor, 9.6% were satisfied with their access to climate information, and only 19.3% rated good and very good.

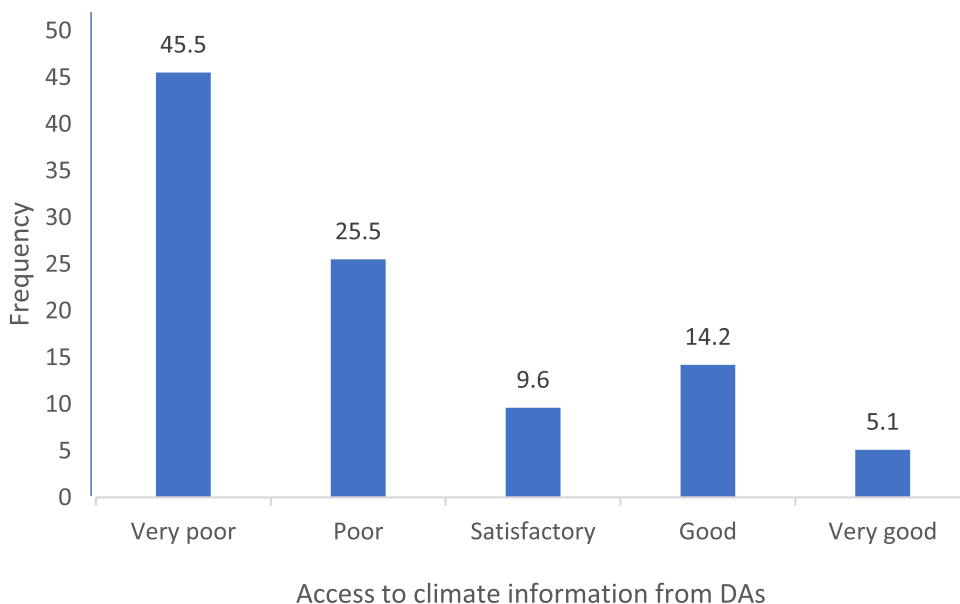


Figure 4. Access to climate change information from the District Assemblies.

Satisfaction with public education on climate change

The study investigated the communities' satisfaction level with public education on climate change. Findings in [Figure 5](#) showed that most (36.6%) of the respondents were satisfied with how they are educated about climate change, with 9.6% feeling very satisfied. Concurrently, while 15.2% were neutral in this context, approximately 14% and 15% were dissatisfied and very dissatisfied, respectively with the climate change education. The

Chi-Square test revealed a statistically significant difference in satisfaction level among the different districts ($\chi^2 = 141.68$; $p = 0.000$).

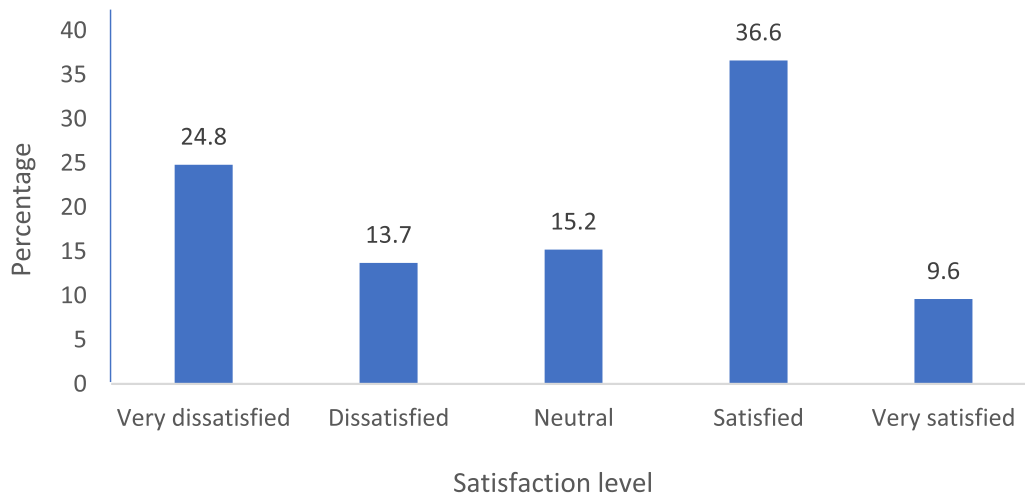


Figure 5. Satisfaction with public climate change education among coastal residents.

Socio-demographic factors as predictors of satisfaction level with public climate change education

This objective sought to find out if the socio-demographic characteristics (gender, age, educational level and community) significantly predict respondents' satisfaction level with climate change education. The hypothesis 'There is no significant influence of socio-demographic factors on the satisfaction level with public climate change education' was formulated to guide the research objective. A multivariate regression was performed to analyse the influence of socio-demographic factors (predictors) on the satisfaction level of climate change education (outcome variable). The regression model adopted was in the form; $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \dots + B_n X_n$ where, Y = respondents' satisfaction level, β_0 , β_1 , β_2 , β_3 , β_4 and $B_n X_n$ are the coefficients of the constant, and the predictors, and X_1 , X_2 , X_3 , X_4 and X_n are the predictors. After satisfying the normality assumption by the normality plot of standardized predicted and standardized residuals, and multi-collinearity by the tolerance and variance inflation factor (VIF), the enter method was employed. The ANOVA results indicated that the regression model was statistically significant in predicting respondents' level of satisfaction with climate change education, $F = 12.903$, $p < 0.05$, $R^2 = 0.159$, R^2 adjusted = 0.147, as shown in [Table 2](#).

The results in [Table 3](#) further indicated that there was a statistically significant influence of the community on respondents' satisfaction level with public education about climate change ($\beta = 0.400$; $p = 0.000$). This implies that as the number of communities increases by one, respondents' satisfaction level is likely to increase by 0.400. Thus, the null hypothesis was rejected and concluded that the community in which one lives influenced respondents' satisfaction level of climate change education. Consequently, there was no statistically significant influence between gender ($\beta = 0.014$; $p = 0.763$), age ($\beta = 0.046$; $p = 0.401$), educational level ($\beta = 0.011$; $p = 0.818$), marital status ($\beta = -0.019$; $p = 0.735$), religion ($\beta = 0.024$; $p = 0.608$) and

Table 2. ANOVA and model summary for demographic factors on satisfaction level with climate education.

Model	Sum of squares	Df	F	Significance	R	R ²	R ² _{Adj}
Regression	4.968	6	12.903	0.000 ^a	0.399 ^b	0.159	0.147
Residual	26.180	408					
Total	31.147	414					

Notes: a. Dependent variable [satisfaction with climate education]. b. Predictors (constant): [gender, age, education, marital status, religion, community].

Table 3. Regression coefficients of socio-demographic factors on satisfaction level with public climate change education.

Variable	<i>B</i>	SE	Beta	<i>T</i>	<i>P</i>	95% CI
Constant	0.326	0.067		4.852	0.000	0.194, 0.458
Gender	0.008	0.026	0.014	0.301	0.763	−0.043, 0.059
Age	0.009	0.011	0.046	0.841	0.401	−0.012, 0.031
Education level	0.002	0.011	0.011	0.231	0.818	−0.019, 0.024
Marital status	−0.005	0.014	−0.019	−0.339	0.735	−0.031, 0.022
Religion	0.007	0.014	0.024	0.513	0.608	−0.021, 0.036
Community	0.088	0.010	0.400	8.725	0.000	0.068, 0.108

Source: Field Data (2023).

satisfaction level with climate change education. Hence, the null hypothesis in each case was accepted. The R^2 adjusted of 0.147 implies that nearly 15% of the variance in climate change education can be explained by the predictors.

Determinants of the effectiveness of climate education

A logistic regression analysis was conducted to examine the factors influencing the effectiveness of climate education (Table 4). The model was statistically significant, $\chi^2(5) = 39.637$, $p < 0.001$, showing that the model's variables contributed substantially in predicting the effectiveness of climate change education. The results revealed that the effect of the type of community was not statistically significant ($\beta = 0.129$; $p = 0.217$), with an odds ratio of $\text{Exp}(\beta) = 1.138$, indicating that the type of community does not significantly affect the likelihood of effective climate education. Contrarily, the type of information channel significantly positively impacted the effectiveness of climate education, with $\beta = 0.183$, $p = 0.015$, $\text{Exp}(\beta) = 1.201$. This indicates that changing the type of information channel is associated with a 20.1% rise in the probability of achieving effective climate education. Similarly, the type of communicator significantly influenced the effectiveness of climate education, $\beta = 0.135$, $p = 0.003$, $\text{Exp}(\beta) = 1.145$, indicating that a change in the type of communicator results in a 14.5% increase in the odds of effective climate education. Trust in the District Assemblies significantly predicted climate education effectiveness, $\beta = 0.640$, $p = 0.021$, $\text{Exp}(\beta) = 1.896$, implying that higher trust in the DAs is associated with nearly a 90% rise in the odds of effective climate education. Attention to climate change information also significantly positively impacted climate education effectiveness, $\beta = 0.386$, $p = 0.003$, $\text{Exp}(\beta) = 1.472$, indicating that increased attention to climate change information is associated with a 47.2% rise in the odds of achieving effective climate education.

Table 4. Factors determining climate change education effectiveness.

Model	<i>B</i>	SE	Wald	Df	Significance	Exp (<i>B</i>)
Type of community	0.129	0.104	1.525	1	0.217	1.138
Type of information channel	0.183	0.130	1.999	1	0.015	1.201
Type of communicator	0.135	0.045	9.136	1	0.003	1.145
Trust in the DAs	0.640	0.277	5.345	1	0.021	1.896
Attention to CCI	0.386	0.129	8.938	1	0.003	1.472
Constant	−1.772	0.730	5.891	1	0.015	0.170

Dependent variable: Effectiveness of climate education.

Discussion

The study highlighted the dual perspectives of coastal residents regarding the role of District Assemblies (DAs) in educating communities about the climate crisis. While the majority of the population strongly agreed that DAs play a critical role in climate education, contrasting views were revealed on the actual educational efforts of DAs over the past 5 years, where the majority perceived a lack of adequate climate education initiatives. This dichotomy underscores a prevalent issue in local governance and community engagement regarding public climate change education. This finding confirms existing studies, indicating that while local government bodies are recognized for their potential impact on climate education, their actual performance often falls

short. According to Bulkeley and Betsill (2013), although local governments translate national climate policies into actionable local plans, however, challenges like limited resources, a lack of technical expertise and insufficient public engagement strategies, impede their effectiveness. Moreover, the gap between recognition and perceived performance implies that community perceptions of local government roles in climate adaptation vary widely, a finding, often influenced by the visibility and tangibility of the initiatives undertaken (International Institute for Environment and Development [IIED], 2019).

The results of poor access to climate change information from the DAs align with previous research indicating that information dissemination on climate change often faces significant barriers in local governance structures (Gupta, 2010). This highlights a critical gap in effective communication strategies by DAs, which, according to Moser and Ekstrom (2010), is essential for fostering community resilience and adaptive capacity to climate change impacts. Respondents' satisfaction level suggests some effectiveness in educational outreach, although it remains relatively modest. The mixed feelings of neutral and dissatisfied or very dissatisfied emphasize the need for enhanced educational efforts tailored to community needs (Sterling et al., 2017). The finding of the community in influencing the satisfaction level with climate change education implies that community-specific factors are vital in determining satisfaction with climate change education in the context of these coastal communities. This could be due to variances in local leadership, resource allocation, environmental awareness and community engagement strategies (Leiserowitz et al., 2005).

Interestingly, the study found no significant influence of gender, age, or education on satisfaction levels, which contrasts with some literature suggesting that demographic factors often shape environmental perceptions and education outcomes (Kollmuss & Agyeman, 2002). This indicates that structural and systemic factors within communities are more pronounced in influencing satisfaction with climate change education than individual demographic characteristics. It is, however, essential to recognize that limited funding and capacity within the DAs can thwart the Assemblies' climate education initiatives. Thus, while calling for localized, context-specific approaches to climate change education, systemic barriers within DAs must be addressed to ensure that communities are engaged and information tailored to these communities for effective climate action at the grassroots level (Pelling, 2010).

The type of media channel and communicator, trust in the DAs, and attention to climate change issues are significant determinants of the effectiveness of climate change education. These findings confirm the literature. The credibility of information sources and the trustworthiness of institutions significantly impact public engagement with climate issues (Pellegrini et al., 2024). Communities are more inclined to act on climate information when it is communicated by trusted and knowledgeable sources (Moser & Dilling, 2011). Complementing this is the type of communication channel – which determines how effectively climate information is received and understood by diverse audiences. Local government should recruit trustworthy communicators and interact with the public via their preferred communication channel. This, according to the Social Learning Theory, will facilitate social learning by creating a platform where communities can share their experiences, learn from others, while observing and modelling environmental conscious behaviours (Bandura, 1977). These behaviours, can spread through social networks, leading to a broader cultural shift toward sustainability. In line with the Theory of Planned Behavior (TPB), public education about climate change can positively influence attitudes and perceived control (Ajzen, 1991), leading to higher engagement in climate-friendly practices. The significant impact of trust in District Assemblies, as revealed in this study, echoes broader findings that institutional trust is essential for fostering public support for climate initiatives.

Theoretical contribution

The theoretical contribution of this study lies in demonstrating how local governance structures – specifically District Assemblies – can act as catalysts for both social learning and behavioural change in the context of climate education. Drawing on Social Learning Theory (SLT), the study shows that when District Assemblies engage trusted communicators and utilize participatory approaches, they create opportunities for observational learning, where community members model climate-conscious behaviours. Similarly, the study advances the TPB by showing that attitudes, perceived social expectations (subjective norms) and perceived behavioural control are shaped not only by individual cognition but also by the credibility of local institutions and

communicators. The theoretical debates on the role local governments play in climate change adaptation was further enriched by shedding light on local governments' potential to strengthen climate-resilient behaviours via educating their constituents about climate change.

Conclusion and policy implications

District Assemblies have existing administrative structures, community outreach mechanisms and mandates for environmental management and education, making them ideal to provide climate education. The findings highlight a notable perception gap concerning the role of District Assemblies in informal climate change education. While over half of the respondents believe that Assemblies are actively involved in such efforts, a majority recognizes a lack of education on climate issues over the past 5 years. This discrepancy suggests that despite perceived involvement, the effectiveness and reach of these educational efforts are inadequate. The gaps identified in the availability and access to climate change information from DAs indicate a pressing need for improved strategies and mechanisms for information dissemination. The satisfaction levels regarding climate change education suggest that while some efforts are effective, there is considerable room for improvement. The finding of the community in influencing this satisfaction implies the need to enhance structural and systemic factors within communities in climate change education than individual socio-demographic factors. Finally, the type of communicator and media channel, trust in DAs, and attention to climate change must inform the development and implementation of Assembly-led climate literacy initiatives to enhance their effectiveness.

This study can inform efforts to enhance education frameworks, suggesting that strengthening local institutions and addressing systemic factors are essential for effective climate action. As nations strive to meet global climate targets, local government actions, guided by informed citizenry, can accelerate sustainable development and resilience. Given this, other jurisdictions should explore how local governments can incorporate climate literacy into their broader development frameworks, ensuring that it becomes a key component of local governance. This requires political will, intergovernmental collaboration, and a clear policy framework that prioritizes climate education at all levels of governance. Multi-stakeholder approach through partnerships between local governments, educational institutions and civil society organizations allows for the co-production of knowledge, blending scientific insights with local experiences to create more relevant and actionable climate education. Countries or regions can replicate this model to enhance community engagement and ensure that climate literacy efforts are grounded in local realities.

Limitations of the study

The adoption of a quantitative approach, while useful for providing general trends in residents' perspective of the District Assemblies in climate literacy education, misses the qualitative aspect, which could provide a nuanced understanding of the problem. Future studies should employ a qualitative approach or integrate both approaches to provide a holistic understanding of the issue. Furthermore, the data collection process was restricted to selected coastal districts and communities in two regions, and largely excluded officials of the District Assemblies, which could have consolidated the findings. This, I consider as a serious limitation that should be considered in future studies.

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Data availability statement

The data associated with this study are those presented in the paper.

Ethical statement

This study is part of MPhil research. Ethical clearance with ID (UCCIRB/CANS/2022/14) was obtained from the University of Cape Coast's Institutional Review Board. In addition, consent was sought from participants before recording the proceedings of the interviews and focus group discussions.

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